

Database System

Quiz 02

Date:

Course code: CS2005

Instructor(s)

Total Time: 15 mins

Mr. Muhammad Naveed

Total marks: 10

Questions: 2

Roll No

Section

Student Signature

Attempt all the questions.

CLO5 Author queries using relational algebra and SQL

Q1: From the following tables (USER and ArRATING), consider the RA query given below. write the final output. (4 marks)

USER

<u>UserID</u>	<u>UserName</u>	Gender
1	Ali	male
2	Ahmed	male
3	Hamza	male
4	kiran	female
5	Maria	female

REPORT

<u>ReportID</u>	Title	Subject	Year
1	Distributed DB for big data	Database	2010
2	Fraud detection using AI	Machine Learning	2023
3	Advanced leftist trees	Data Structure	2001
4	Applications of Neural Networks	Machine Learning	1994
5	AVL Trees	Data Structure	2000

REPORT_AUTHOR

<u>ReportID</u>	<u>AuthorID</u>
1	1
1	2
2	3
3	4
4	5
5	1

RATING

<u>ReportID</u>	<u>UserID</u>	UserRating	UserReview
1	1	4	Excellent
1	2	5	Highly recommended
3	2	3	Good read
4	4	4	Classic literature
5	5	2	Not my cup of tea

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Subject $\propto$ COUNT(Subject) (RATING * REPORT)

Output

Subject	count
Database	5
Machine Learning	10
Data Structure	10

Q2: Write SQL and RA statements to find usernames of users who rated more than one report. 6 marks

Sql:

```
SELECT u.UserName FROM USER u JOIN RATING r ON u.UserID = r.UserID GROUP BY u.UserID,  
u.UserName HAVING COUNT(DISTINCT r.ReportID) > 1;
```

```
SELECT UserName FROM USER WHERE UserID IN ( SELECT UserID FROM RATING GROUP BY UserID  
HAVING COUNT(DISTINCT ReportID) > 1);
```

RA:

```
T1  $\leftarrow$   $\gamma$  UserID, COUNT(ReportID)  $\rightarrow$  c (RATING)
```

```
T2  $\leftarrow$   $\sigma$  c > 1 (T1)
```

```
Result  $\leftarrow$   $\pi$  UserName (T2  $\bowtie$  USER)
```